

Phase 1: Fundamentals (Days 1-10)	1
Phase 2: Deep Learning and Transformers (Days 11-20)	2
Phase 3: Generative AI and Large Language Models (LLMs) (Days 21-30)	2
Phase 4: LLM Applications and Fine-Tuning (Days 31-40)	3
Phase 5: Retrieval-Augmented Generation (RAG) (Days 41-50)	4
Phase 6: LLM Evaluations (Days 51-55)	4
Phase 7: AI Ethics and Governance (Days 56-60)	5

Phase 1: Fundamentals (Days 1-10)

Objective: Establish a strong foundation in Python programming, Machine Learning (ML), and Neural Networks (NN).

Topics:

- 1. Python for AI:**
 - Data manipulation with NumPy and Pandas.
 - Data visualization using Matplotlib.
 - Setting up and utilizing Jupyter Notebooks.
- 2. Machine Learning Basics:**
 - Understanding supervised and unsupervised learning.
 - Implementing algorithms like Linear and Logistic Regression.
- 3. Neural Networks (NN):**
 - Comprehending perceptrons and feedforward networks.

Activities:

- Complete Python tutorials focusing on data science applications.
- Train basic ML models using Scikit-learn.
- Build simple neural networks with TensorFlow or PyTorch.

Online Resources:

- **Python:**
 - [Automate the Boring Stuff with Python](#)
 - [Python Data Science Handbook](#)
- **Machine Learning:**
 - [Google's Machine Learning Crash Course](#)
 - [Introduction to Machine Learning with Python](#)
- **Neural Networks:**

- [Neural Networks and Deep Learning](#)
 - [Deep Learning with Python](#)
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Phase 2: Deep Learning and Transformers (Days 11-20)

Objective: Gain a comprehensive understanding of deep learning architectures and transformer models.

Topics:

1. **Deep Learning (DL):**
 - Convolutional Neural Networks (CNNs) for image processing.
 - Recurrent Neural Networks (RNNs) for sequence data.
2. **Transformers:**
 - Mechanics of self-attention.
 - Architectures like BERT and GPT.

Activities:

- Implement CNNs and RNNs on datasets such as MNIST and IMDB.
- Fine-tune a pre-trained transformer model using the Hugging Face library.

Online Resources:

- **Deep Learning:**
 - [Deep Learning Specialization by Andrew Ng](#)
 - [CS231n: Convolutional Neural Networks for Visual Recognition](#)
 - **Transformers:**
 - The Illustrated Transformer
 - [Attention Is All You Need](#)
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Phase 3: Generative AI and Large Language Models (LLMs) (Days 21-30)

Objective: Explore generative models and delve into the workings of LLMs.

Topics:

1. **Generative AI:**
 - Generative Adversarial Networks (GANs).
 - Variational Autoencoders (VAEs).
2. **Large Language Models (LLMs):**
 - Architecture and training methodologies of models like GPT, T5, and BERT.

- Techniques in prompt engineering.

Activities:

- Develop a simple GAN for image generation tasks.
- Train a small-scale LLM using the Hugging Face Transformers library.

Online Resources:

- **Generative AI:**
 - [Ian Goodfellow's Deep Learning Book](#)
 - [Generative Adversarial Networks \(GANs\) – A Comprehensive Guide](#)
 - **LLMs:**
 - [Hugging Face Course](#)
 - [The Illustrated GPT-2](#)
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Phase 4: LLM Applications and Fine-Tuning (Days 31-40)

Objective: Apply LLMs to real-world scenarios and learn fine-tuning techniques for specific tasks.

Topics:

1. **LLM Applications:**
 - Developing chatbots, text summarization tools, and translation systems.
2. **Fine-Tuning:**
 - Adapting models on domain-specific data using platforms like OpenAI, Hugging Face, or LLaMA.

Activities:

- Create an LLM-powered chatbot utilizing OpenAI's API.
- Fine-tune a T5 or GPT model for tasks such as question answering.

Online Resources:

- **LLM Applications:**
 - [LangChain Documentation](#)
 - [Building a Chatbot with OpenAI's GPT-3](#)
- **Fine-Tuning:**
 - [Hugging Face Fine-Tuning Tutorial](#)
 - [Fine-Tuning GPT-3 for Specific Tasks](#)

Phase 5: Retrieval-Augmented Generation (RAG) (Days 41-50)

Objective: Master advanced techniques in Retrieval-Augmented Generation to enhance LLM capabilities.

Topics:

1. RAG Fundamentals:

- Understanding the integration of retrieval systems with LLMs to provide contextually relevant information.

2. Advanced RAG Techniques:

- Implementing query expansion methods, such as BM25, to improve retrieval accuracy.
- Utilizing synthetic data to augment training datasets.
- Optimizing indexing strategies for efficient data retrieval.

Activities:

- Develop a RAG pipeline using OpenAI embeddings and ChromaDB.
- Apply BM25 for query expansion to enhance information retrieval.

Online Resources:

● RAG Tutorials:

- [A Beginner's Guide to Building a Retrieval Augmented Generation \(RAG\) Application from Scratch](#)
- [Mastering BM25: A Deep Dive into the Algorithm and Its Application in Milvus](#)

● Advanced RAG Techniques:

- [Advanced RAG Techniques: Elevating Your Retrieval-Augmented Generation Systems](#)
 - [Query Expansion in RAG: Enhancing Retrieval Accuracy](#)
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Phase 6: LLM Evaluations (Days 51-55)

Objective: Learn to assess the performance and reliability of Large Language Models.

Topics:

1. Evaluation Metrics:

- Understanding metrics such as BLEU, ROUGE, and perplexity.

2. Human Evaluation:

- Conducting assessments for coherence, relevance, and factual accuracy.

3. Bias and Fairness:

- Identifying and mitigating biases in model outputs.

Activities:

- Evaluate an LLM's performance on a specific task using both automated metrics and human judgment.
- Analyze model outputs for potential biases and propose mitigation strategies.

Online Resources:

- **Evaluation Metrics:**
 - Understanding Evaluation Metrics for Language Models
 - [A Survey of Evaluation Metrics Used for NLG Systems](#)
 - **Bias and Fairness:**
 - [Mitigating Bias in AI Models](#)
 - [Fairness in Machine Learning: A Survey](#)
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Phase 7: AI Ethics and Governance (Days 56-60)

Objective: Understand the ethical considerations and governance frameworks essential for responsible AI development.

Topics:

1. **Ethical Principles:**
 - Fairness, accountability, transparency, and privacy in AI systems.
2. **Regulatory Frameworks:**
 - Overview of global AI regulations and standards.
3. **Responsible AI Practices:**
 - Implementing ethical guidelines in AI development and deployment.

Activities:

- Analyze case studies of AI applications to identify ethical challenges and propose solutions.
- Develop a set of ethical guidelines for an AI project.

Online Resources:

- **AI Ethics:**
 - [The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems](#)
 - AI Ethics Guidelines Global Inventory
- **Responsible AI:**
 - Responsible AI Practices by Google
 - [Microsoft's AI Principles](#)